

Department of Design

Indian Institute of Technology Delhi

Registration for Industry / Research Design Project (DSD 892)

Objective of this Project: To develop the ability to look at design problems from a research perspective. The student is expected to contribute to the professional design field through new design knowledge generation. The **Graduation Project** is aimed to polish the designer's research skills. The designer is expected to deliver cutting-edge research and be able to articulate it professionally.

Name of the Student: Sanket Subhash Naik

Entry Number: 2020DDS6012

Project Supervisors from IIT Delhi (IIT):

Guide: Prof. Gourab Kar

Co-guide: Prof. Sabyasachi Paldas

Safer washrooms for the elderly

Project brief

Elderly people have slips and falls in the washroom, which can lead to long term injuries and can even be fatal. Many elderly seek independence and privacy specially in the washroom. The washroom space at home needs to be redesigned considering the needs of the elderly, many of whom suffer from challenges such as reduced vision, low dexterity, reduced muscle strength, lower balance etc. I would like to come up with a range of products, practices, and guidelines to make the washroom safer for the elderly.

Project scope

The project would incorporate the following:

- Research into falls and injuries in the toilets and bathroom (Washrooms).
- Research in aspects concerning washroom design such as building structure, layouts, plumbing, flooring, wiring, lighting, HVAC, cleaning, and sanitization, etc.
- Ethnography to find out the needs of the elderly in the washroom.
- Prototyping solutions that increase safety and usability for elderly in the washroom.
- Testing solutions to validate improvements in safety and usability for elderly in the washroom.
- Communicating design solutions, safety and sanitation practices and guidelines to create safer washrooms for the elderly.

Timeline

<i>Week 1 - Week 3 :</i>	Research, Presentation of initial concepts
<i>Week 4 - Week 6 :</i>	Ideation/conceptualization and scaled models
<i>Week 7 :</i>	Full scale model of washroom space
<i>Week 8 - Week 12 :</i>	Prototyping and testing
<i>Week 13 - Week 18 :</i>	Development of final prototypes
<i>Week 19 - Week 22 :</i>	Project documentation, Testing and improvement

Expected deliverables

- Prototypes with multiple rounds of user testing and iteration.
- Document providing guidelines for making safer washrooms for the elderly.
- Presentation and Final thesis report document.



Student's signature
Sanket Subhash Naik



Guide's signature
Gourab Kar



Co-Guide's signature
Sabyasachi Paldas